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CS 499 Computer Science Capstone

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CS 499 Milestone Four: Enhancement Three: Databases

Continuing onto the enhancement of databases, artifact is the Android Inventory App developed in CS 360 that utilizes a local SQLite database that I have transitioned into a MEAN full stack web application called InvenX, which uses a MongoDB non-relational database. The original Android version developed in Java allowed users to manage inventory items on their phone, locally, with no extension online or across other platforms. After transforming the applications' foundation into a full-stack web application, the application uses MongoDB database, which allows access over the internet and across devices, ideally. Obviously, the development is only on my local platform, and not being hosted actively.

My choice for the artifact is because it shows versatility for not only development of a full stack application with a hosted database, but also the ability to transition from SQL language to MongoDB for CRUD operations with the potential for having scalable web app versus local hosting on a phone based app. While I do find the general idea and structures to be similar, I've become better at handling MongoDB queries using Mongoose in place of SQL language. During the process of development, I integrated the backend API with Angular and have now included the ability to filter based on range sliding bars and checkboxes, search based on user input text fields, and radio button oriented sort functions. All are functional through real-time manipulation of the item dataset. I would say that my enhancement with working on databases shows my understanding of the design of databases and the ability to manage a backend database in

conjunction with connecting to the frontend, while securing data between the two and preventing injection attacks through sanitization.

Comparing with my initial plan in Module One, I would argue that I am around 85 to 90% completed for the database enhancement except for ensuring security and coordinating how the data is displayed on the UI. The database structure and operations, to the best of my knowledge, are fully functioning and complete. All CRUD endpoints work correctly, and I've verified that inventory updates reflect instantly in the UI. My frontend filtering and searching is implemented, but I am still unsure if my current implementation of these meets the goals of the previous enhancement for algorithms, as I simply utilized the `sort()` function. My target for the next week will be to re-evaluate the current implementation of the sorting algorithm, possibly creating my own, more than likely merge sort, and then improvements in UI/UX for the angular frontend. In short, I'm not using advanced algorithms yet, but I have created a filter.

The artifact enhancement for databases has taught me more about the connection between the frontend UI with the database and how each reflects what is happening in the backend. For example, making sure that the sorting state in Angular matches the data that's being rendered and that any changes in the database result in real-time updates to the interface. My biggest challenge has been putting together the filtering logic for search, category selection, ranges, and sort direction with each needing to interact correctly. I still think I am a ways off on completion, I still am unsure on algorithms enhancement, as I previously stated, but as far as the database enhancement is concerned, I think I have this portion completed.